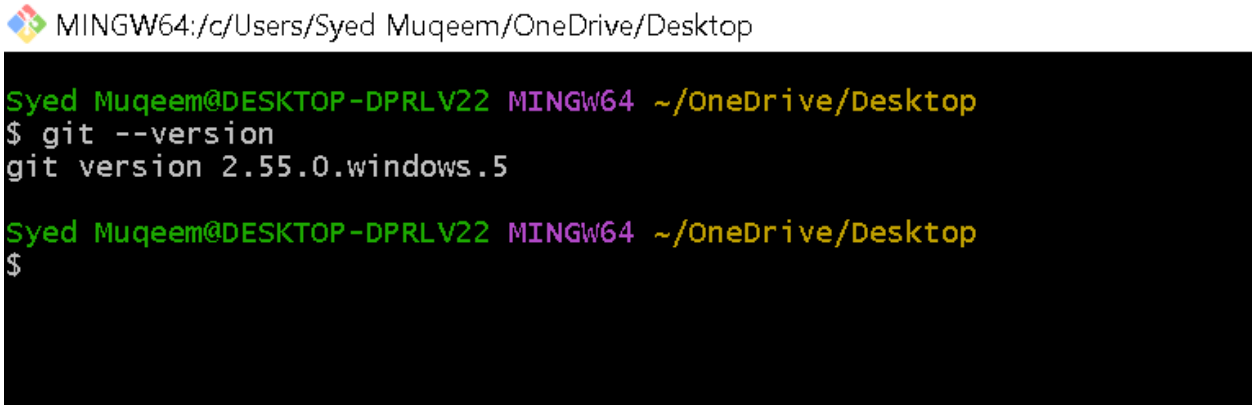
Task-2 on_Github

1. Install git.

Git has been installed already, to check the version we use the below command

: git --version



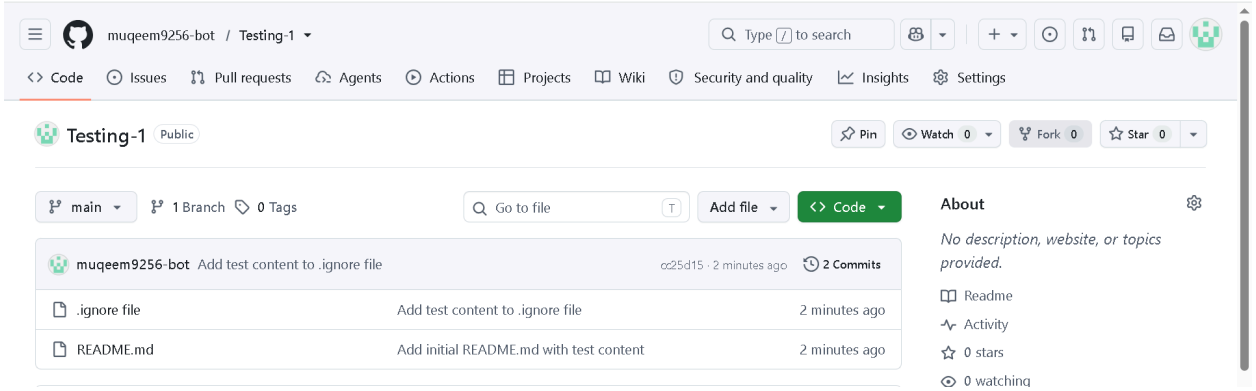
2. Create a repo in github with README.md and .ignore file.

Steps to create repositories:

Step1: Click on add repository, enter the details.

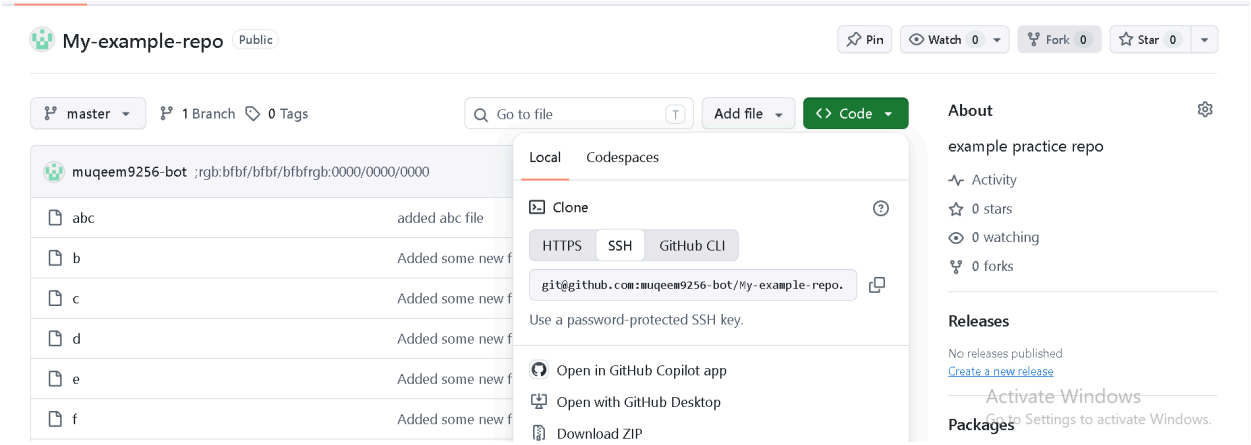
Step2: Click on create repository

Step3: click on add new file, enter file name and commit



3. Clone the created repo to local.

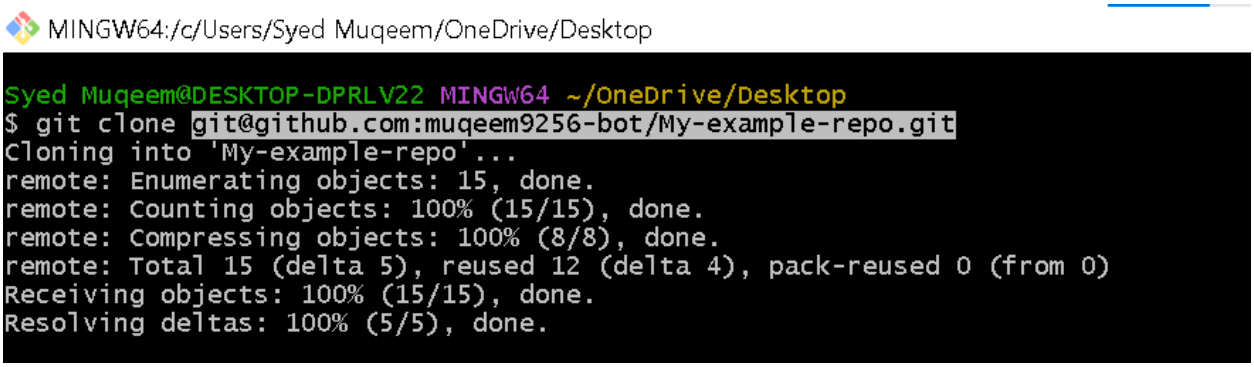
Step1 go to the repo and click on “code” dropdown copy the ssh URL.



Step2: go to git bash write the command and paste the ssh URL.

: git clone git@github.com:muqeem9256-bot/My-example-repo.git

you have successfully downloaded the repository from remote to local.



4. Create two files in local repo.

#To create a file in the local repository use the below command

: touch file-1 file-2

if you ls anc check you will have tow files created in the local repo.

```
Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/local-repo
$ git init
Initialized empty Git repository in C:/Users/Syed Muqeem/OneDrive/Desktop/local-repo/.git/

Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/local-repo (master)
$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        file-1
        file-2

nothing added to commit but untracked files present (use "git add" to track)

Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/local-repo (master)
$
```

5. Commit two files and push to the central Repository.

```
Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/local-repo (master)
$ git commit -m "added file-1 and file-2 in master branch"
[master (root-commit) f5a2569] added file-1 and file-2 in master branch
 2 files changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 file-1
 create mode 100644 file-2
```

To push files from your local to central repo, you need to add your GitHub repository as a remote. To check remote repo use command : **git remote -v**

to add the remote repo use the command

: git remote add origin origingit@github.com:muqeem9256-bot/Testing-1.git

MINGW64:/c/Users/Syed Muqeem/OneDrive/Desktop/local-repo

```
Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/local-repo (master)
$ git remote add origin git@github.com:muqeem9256-bot/Testing-1.git

Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/local-repo (master)
$ git remote -v
origin  git@github.com:muqeem9256-bot/Testing-1.git (fetch)
origin  git@github.com:muqeem9256-bot/Testing-1.git (push)

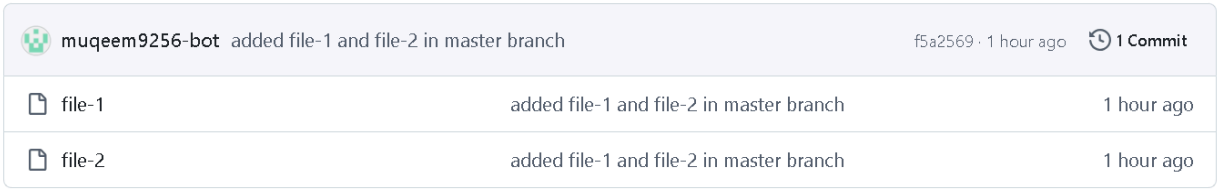
Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/local-repo (master)
$
```

#Now use the command git push to push the files to the central repository.

: git push -u origin master

```
Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/local-repo (master)
$ git push -u origin master
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 230 bytes | 230.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
To github.com:muqeem9256-bot/Testing-1.git on GitHub by visiting:
 * [new branch]      master -> masterm9256-bot/Testing-1/pull/new/master
branch 'master' set up to track 'origin/master'.

Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/local-repo (master)
$
```



6. Create a branch in local and create a sample file and push to central.

To create a branch we use the command : git branch <branch name>

: git branch branch-1

To list the available branches we use the command

: git branch - -list

To switch the branch we use the command : git checkout <branch name>

: git checkout branch-1

```
Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/local-repo (master)
$ git branch branch-1

Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/local-repo (master)
$ git branch --list
  branch-1
* master

Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/local-repo (master)
$ git checkout branch-1
Switched to branch 'branch-1'

Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/local-repo (branch-1)
$ touch file-1
```

Now pushing the new created sample file-1 and file-3 to the central repo.

```
Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/local-repo (branch-1)
$ ls
file-1  file-3

Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/local-repo (branch-1)
$ git status
On branch branch-1
Untracked files:
  (use "git add <file>..." to include in what will be committed)
        file-3

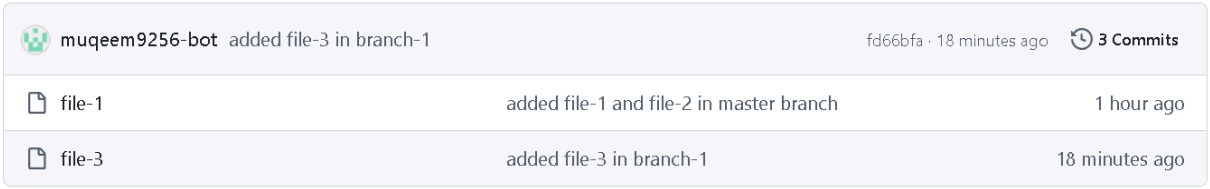
nothing added to commit but untracked files present (use "git add" to track)

Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/local-repo (branch-1)
$ git add .

Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/local-repo (branch-1)
$ git status
On branch branch-1
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        new file:   file-3


Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/local-repo (branch-1)
$ git commit -m "added file-3 in branch-1"
[branch-1 fd66bfa] added file-3 in branch-1
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 file-3

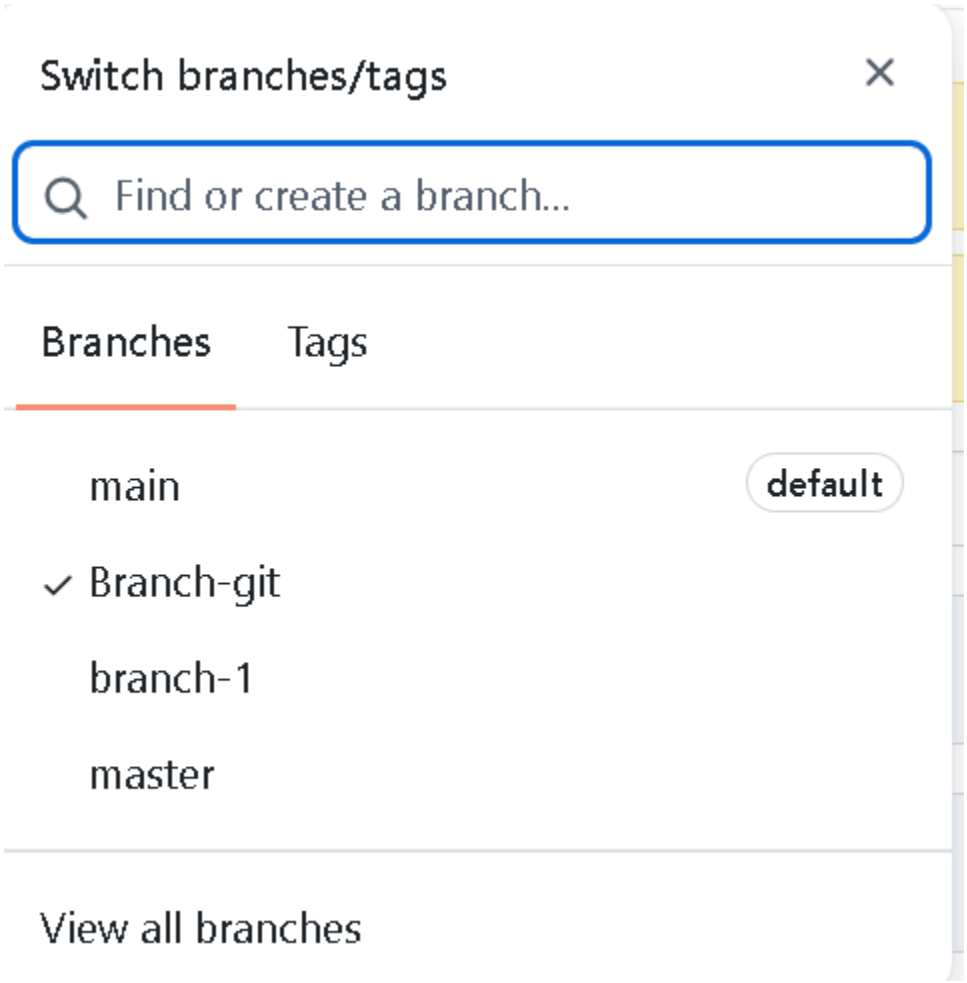
Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/local-repo (branch-1)
$ git push -u origin branch-1
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 2 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (4/4), 449 bytes | 449.00 KiB/s, done.
Total 4 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'branch-1' on GitHub by visiting:
remote:      https://github.com/muqeem9256-bot/Testing-1/pull/new/branch-1
remote:
        To github.com:muqeem9256-bot/Testing-1.git
 * [new branch]      branch-1 -> branch-1
branch 'branch-1' set up to track 'origin/branch-1'.
```



7. Create a branch in github and clone that to local.

To create a branch in git hub go to—-**branches** button and click on —**new branch** button, enter the branch name and click— create branch

I created **Branch-git** as a new branch



To clone a branch to the local repository we use the command

: git clone -b Branch-git git@github.com:muqeem9256-bot/Testing-1.git

```
Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/local-repo (master)
$ ls
file-1  file-2

Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/local-repo (master)
$ git clone -b Branch-git git@github.com:muqeem9256-bot/Testing-1.git
Cloning into 'Testing-1'...
remote: Enumerating objects: 16, done.
remote: Counting objects: 100% (16/16), done.
remote: Compressing objects: 100% (11/11), done.
remote: Total 16 (delta 1), reused 7 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (16/16), done.
Resolving deltas: 100% (1/1), done.

Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/local-repo (master)
$ ls
Testing-1/  file-1  file-2

Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/local-repo (master)
$
```

in order to check the branch i have created in github is available in the local we need to go that downloaded repository '**cd Testing-1**'

#here we can see the branch name has beed changed to **Branch-git**

```
Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/local-repo (master)
$ cd Testing-1

Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/local-repo/Testing-1 (Branch-git)
$ git branch --list
* Branch-git

Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/local-repo/Testing-1 (Branch-git)
$
```

8. Merge the created branch with master in git local.

#To merge the branch with another branch we have to first go the the destination branch and the use the command below.

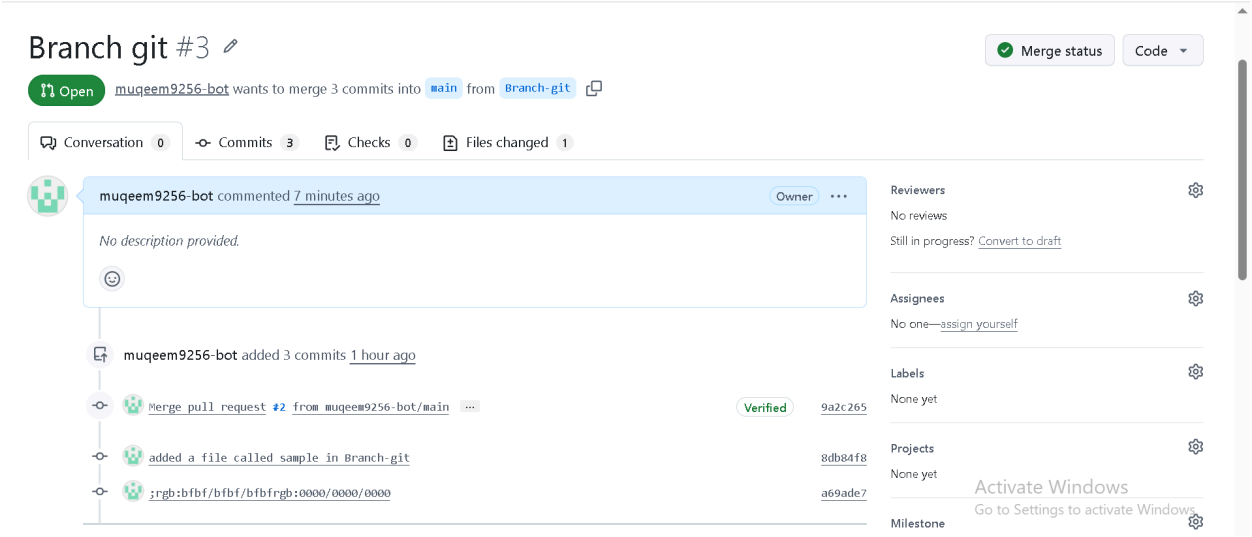
: git merge Branch-git

It has been merged now i can see the files from the **Branch-git** in **master branch.**

```
Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/local-repo/Testing-1 (master)
$ ls
README.md  file-1  file-2  git-file1

Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/local-repo/Testing-1 (master)
$
```

9. Merge the created branch with master in github by sending a pull request.

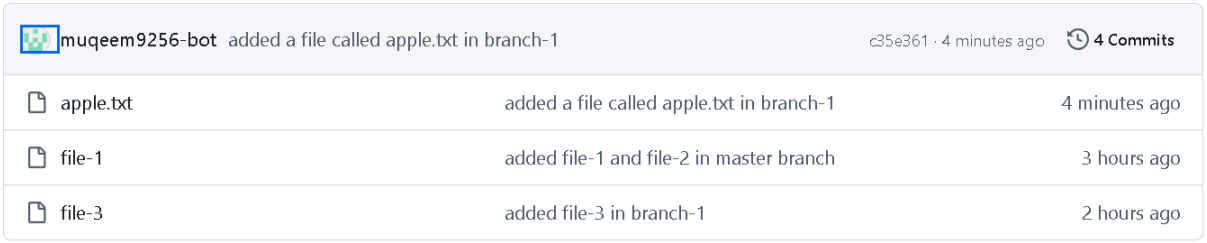


10. **Create a file in local and send that to the branch in github**.

Created a file with name apple.txt in the local branch-1 and added to tracking area

And committed with **"added a file called apple.txt in branch-1"** and pushed it to the branch in git hub.

you can see the file apple.txt is available in the github branch-1



```
Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/local-repo (branch-1)
$ touch apple.txt

Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/local-repo (branch-1)
$ git status
On branch branch-1
Your branch is up to date with 'origin/branch-1'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        apple.txt

nothing added to commit but untracked files present (use "git add" to track)

Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/local-repo (branch-1)
$ git add .

Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/local-repo (branch-1)
$ git status
On branch branch-1
Your branch is up to date with 'origin/branch-1'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        new file:   apple.txt


Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/local-repo (branch-1)
$ git commit -m "added a file called apple.txt in branch-1"
[branch-1 c35e361] added a file called apple.txt in branch-1
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 apple.txt

Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/local-repo (branch-1)
$ git push -u origin branch-1
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (2/2), 263 bytes | 263.00 KiB/s, done.
Total 2 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:muqeem9256-bot/Testing-1.git
   fd66bfa..c35e361  branch-1 -> branch-1
branch 'branch-1' set up to track 'origin/branch-1'.
```

11. Clone only a branch from github to local.

```
Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/local-repo (branch-1)
$ git clone -b Apple git@github.com:muqeem9256-bot/Testing-1.git
Cloning into 'Testing-1'...
remote: Enumerating objects: 23, done.
remote: Counting objects: 100% (23/23), done.
remote: Compressing objects: 100% (16/16), done.
remote: Total 23 (delta 3), reused 13 (delta 2), pack-reused 0 (from 0)
Receiving objects: 100% (23/23), done.
Resolving deltas: 100% (3/3), done.

Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/local-repo (branch-1)
$ ls
Testing-1/  apple.txt  file-1  file-3

Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/local-repo (branch-1)
$ git pull
remote: Enumerating objects: 5, done.
remote: Counting objects: 100% (5/5), done.
remote: Compressing objects: 100% (1/1), done.
remote: Total 4 (delta 2), reused 4 (delta 2), pack-reused 0 (from 0)
Unpacking objects: 100% (4/4), 397 bytes | 9.00 KiB/s, done.
From github.com:muqeem9256-bot/Testing-1
 * [new branch]      Apple      -> origin/Apple
   9a2c265..a69ade7  Branch-git -> origin/Branch-git
Already up to date.

Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/local-repo (branch-1)
$ cd Testing-1/

Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/local-repo/Testing-1 (Apple)
$
```

12. Create a file with all passwords and make that untrackable with git.

```
Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/local-repo (main)
$ vi secret.txt

Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/local-repo (main)
$ touch .gitignore

Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/local-repo (main)
$ echo "secret.txt" >> .gitignore

Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/local-repo (main)
$ git status
On branch main
Your branch is up to date with 'origin/main'.

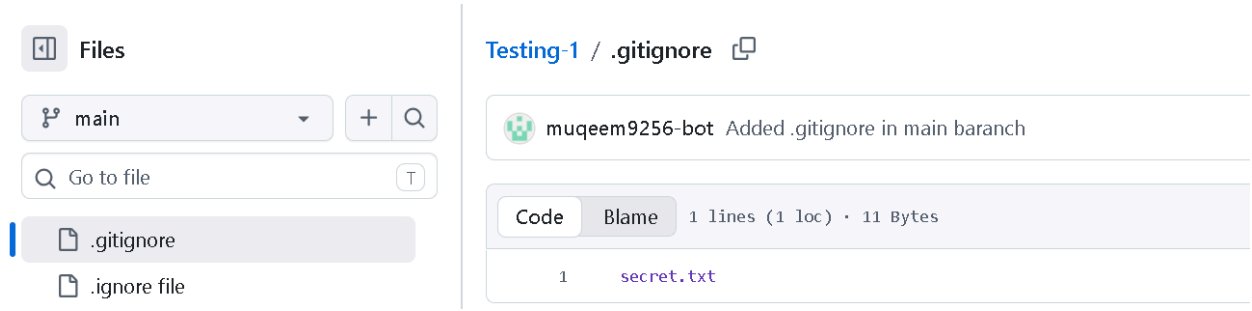
Untracked files:
  (use "git add <file>..." to include in what will be committed)
        .gitignore

nothing added to commit but untracked files present (use "git add" to track)

Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/local-repo (main)
$ git add .
warning: in the working copy of '.gitignore', LF will be replaced by CRLF the next time Git touches it

Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/local-repo (main)
$ git commit -m "Added .gitignore in main baranch"
[main 33bfb00] Added .gitignore in main baranch
 1 file changed, 1 insertion(+)
 create mode 100644 .gitignore

Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/local-repo (main)
$ git push
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 361 bytes | 361.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:muqeem9256-bot/Testing-1.git
   6223876..33bfb00  main -> main
```



13. Make a commit and make that commit reset without saving changes.

Step 1: Create a file

Step 2: Add and commit the file

Step 3: Reset the commit and discard changes

Step 4: Check the result

Below you can see the result with all steps

```
Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/local-repo (main)
$ touch apple

Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/local-repo (main)
$ git status
On branch main
Your branch is up to date with 'origin/main'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        apple

nothing added to commit but untracked files present (use "git add" to track)

Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/local-repo (main)
$ git add .

Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/local-repo (main)
$ git commit -m "added a file apple in main branch"
[main a206eea] added a file apple in main branch
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 apple

Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/local-repo (main)
$ git reset --hard HEAD~1
HEAD is now at 33bfb00 Added .gitignore in main baranch

Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/local-repo (main)
$ ls
README.md  git-file1  secret.txt

Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/local-repo (main)
$
```

14. Revert a committed commit to the older version.

Step 1: Create a file

Step 2: Add and commit the file

Step 3: Reset the commit and discard changes

Step 4: Check the result

```
Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/local-repo (main)
$ touch apple

Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/local-repo (main)
$ git status
On branch main
Your branch is up to date with 'origin/main'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        apple

nothing added to commit but untracked files present (use "git add" to track)

Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/local-repo (main)
$ git add .

Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/local-repo (main)
$ git commit -m "added file apple in main branch"
[main c743dbd] added file apple in main branch
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 apple

Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/local-repo (main)
$ git log --oneline
c743dbd (HEAD -> main) added file apple in main branch
33bfb00 (origin/main, origin/HEAD) Added .gitignore in main baranch
6223876 (origin/Apple) Merge pull request #1 from muqeem9256-bot/Branch-git
11dae7c Create sample file in GitHub branch
cc25d15 Add test content to .ignore file
a7d09e1 Add initial README.md with test content
```

MINGW64:/c/Users/Syed Muqeem/OneDrive/Desktop/local-repo

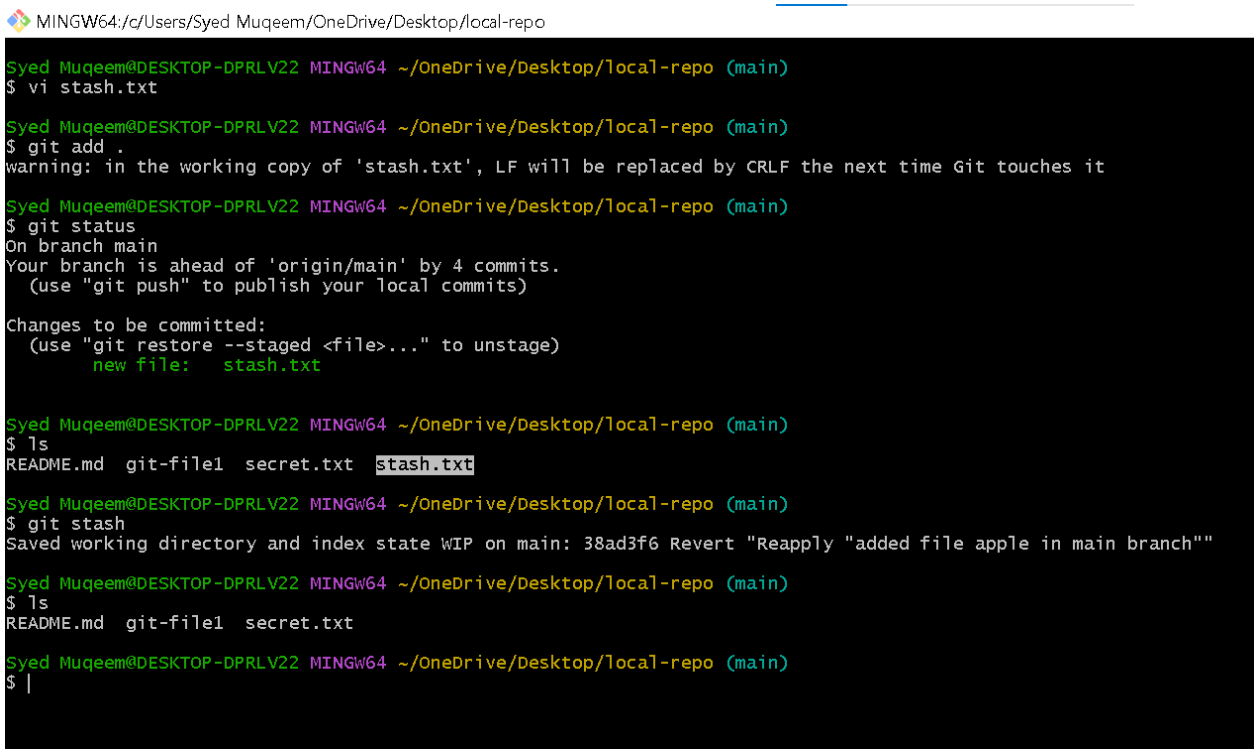
```
;rgb:bfbf/bfbf/bfbfrgb:0000/0000/0000

This reverts commit ed34a487c5edde1ac891fc163e34b5698d5d0f43.

# Please enter the commit message for your changes. Lines starting
# with '#' will be ignored, and an empty message aborts the commit.
#
# On branch main
# Your branch is ahead of 'origin/main' by 2 commits.
#   (use "git push" to publish your local commits)
#
# Changes to be committed:
#       new file:   apple
#
~
```

```
Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/local-repo (main)
$ git log --oneline
38ad3f6 (HEAD -> main) Revert "Reapply "added file apple in main branch""
c2198ea Reapply "added file apple in main branch"
ed34a48 Revert "added file apple in main branch"
c743dbd added file apple in main branch
33bfb00 (origin/main, origin/HEAD) Added .gitignore in main baranch
6223876 (origin/Apple) Merge pull request #1 from muqeem9256-bot/Branch-git
11dae7c Create sample file in GitHub branch
cc25d15 Add test content to .ignore file
a7d09e1 Add initial README.md with test content
```

15. Push a file to stash without saving the changes and work on another file.

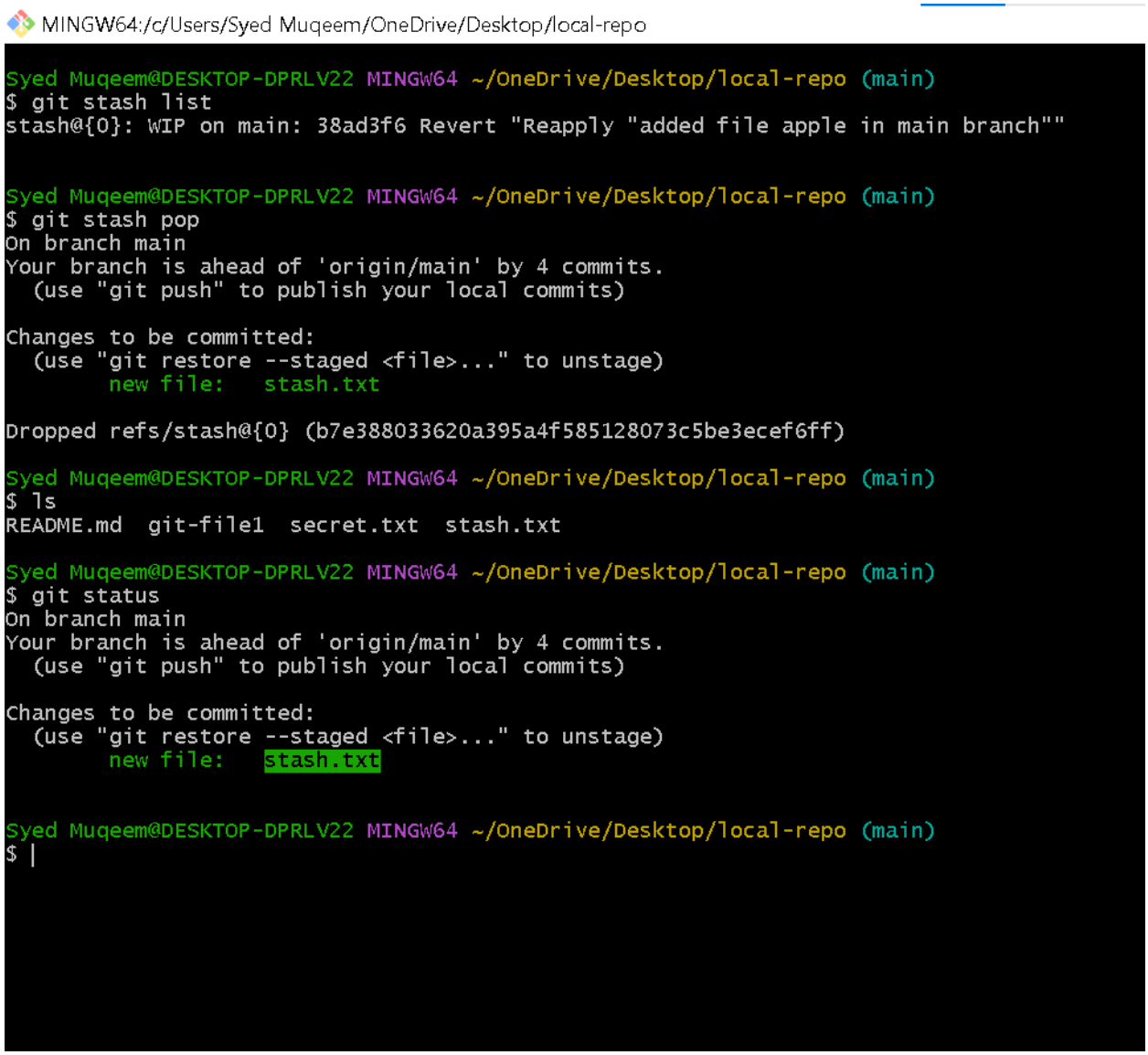


16. Undo the stash file and start working on that again.

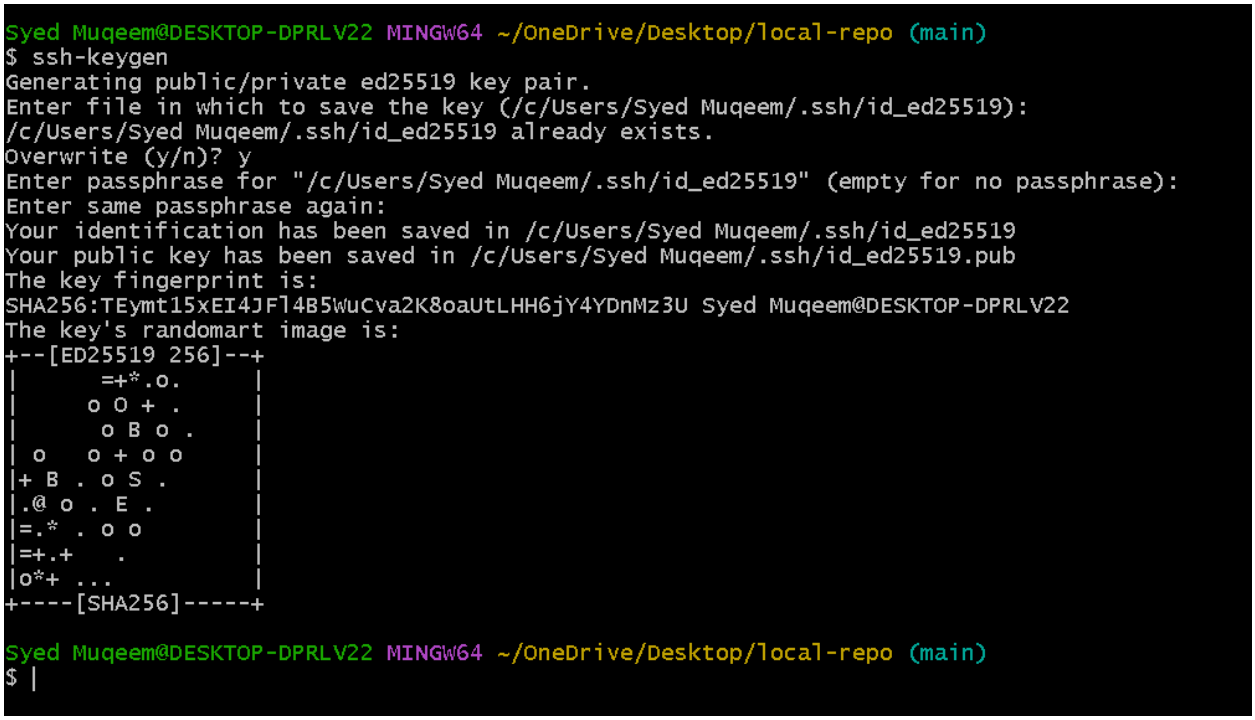
Step 1: Check your stashed changes

Step 2: Restore the changes

Step 3: Check your changes



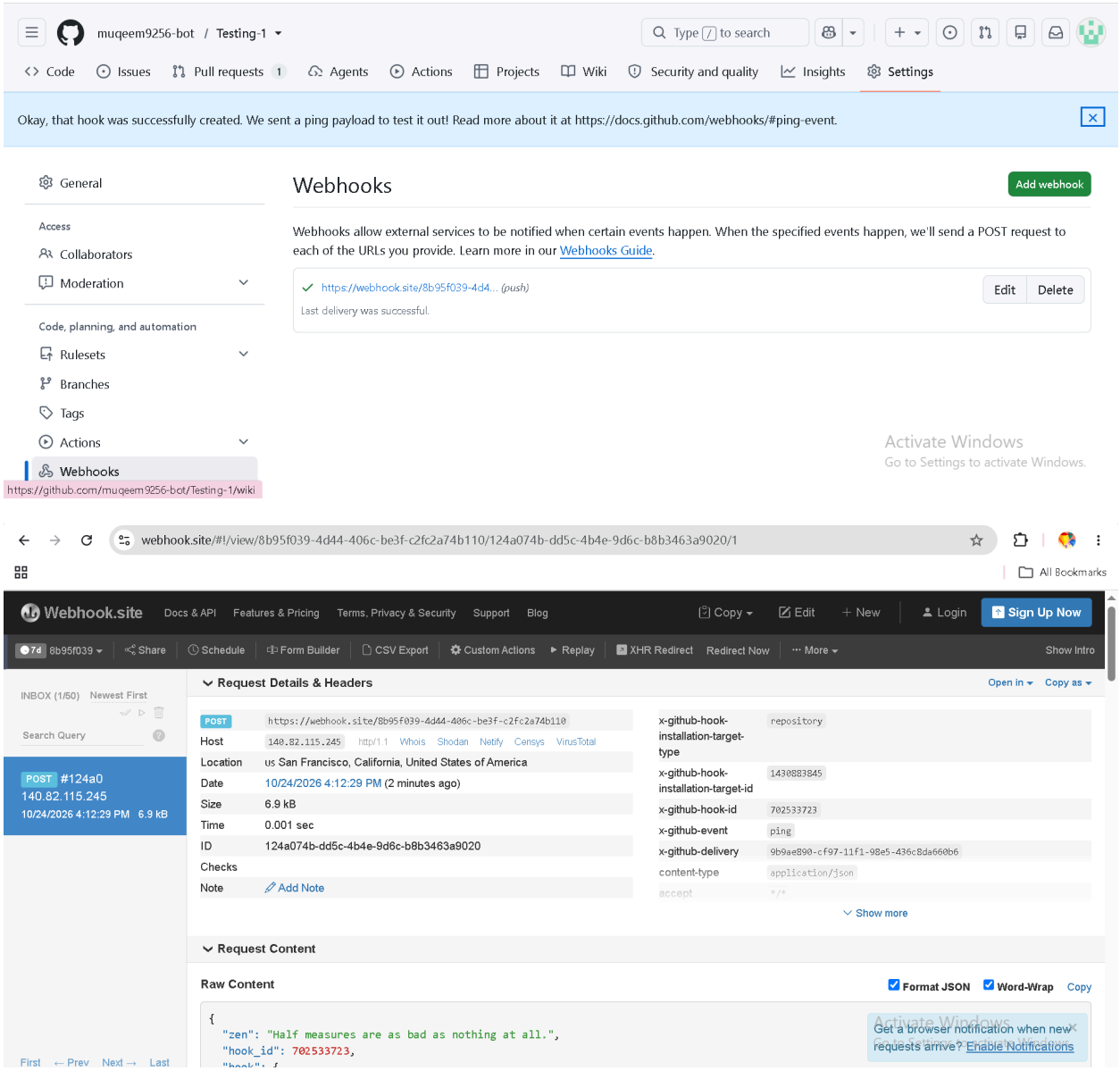
17. Generate a ssh-keygen and configure it into github.



Add to GitHub:

- Go to GitHub > Settings > SSH and GPG Keys > New SSH Key
- Paste the copied key
- Give it a title and Save

18. Configure webhooks to github.

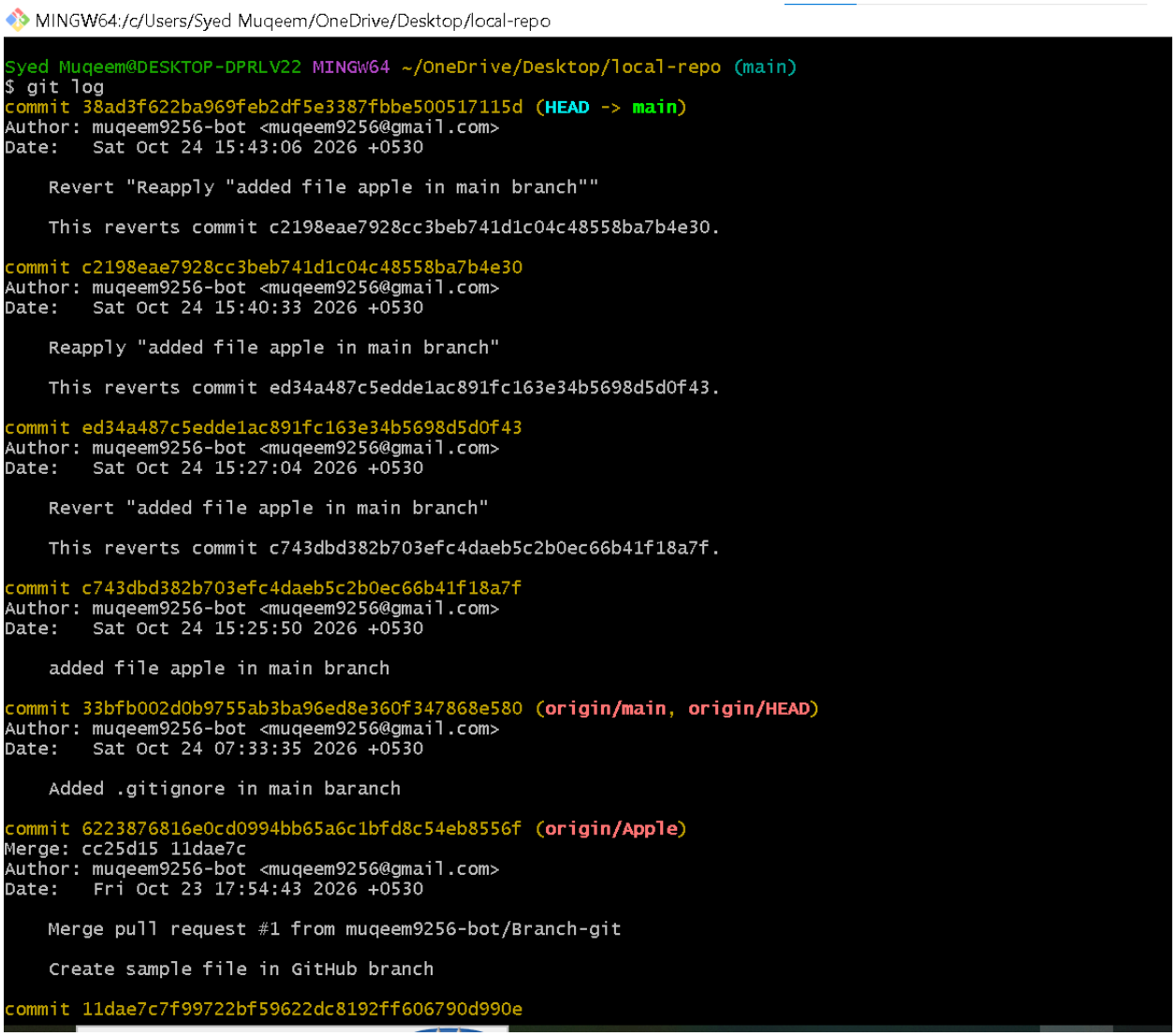


19. Basic understanding of .git file.

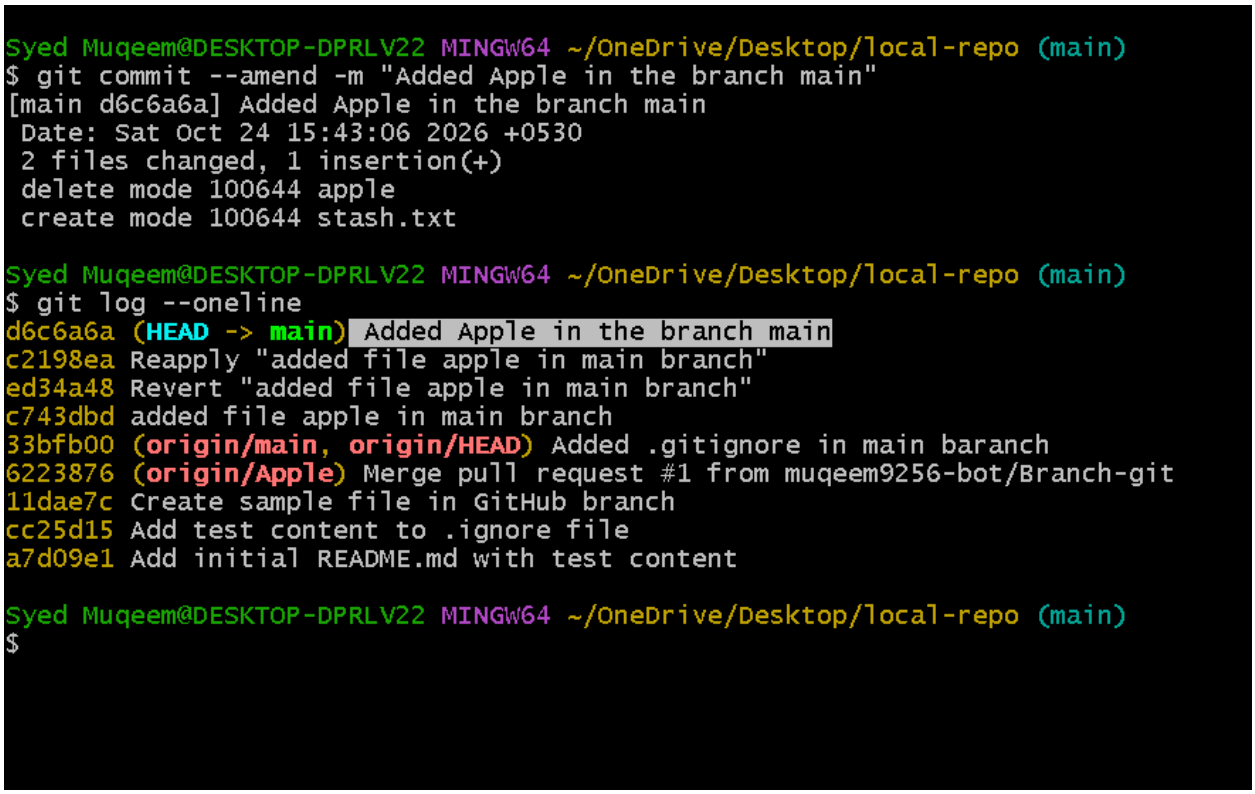
1: The .git folder is a hidden folder that stores all the important information Git needs to track your project.

2: When you initialize a repo (git init), Git creates a hidden .git directory in your project.

20. Check all the logs of git.



21. Rename the commit message.



22. Merge multiple commits into single commits.

Step 1: Check your commit history

#Suppose I want to combine the latest 3 commits into one

Step 2: Start interactive rebase

A text editor will open with something “pick” Changed 2 of them to “squash”

MINGW64:/c/Users/Syed Muqeem/OneDrive/Desktop/local-repo

```
pick ed34a48 # Revert "added file apple in main branch"
squash c2198ea # Reapply "added file apple in main branch"
squash d6c6a6a # Added Apple in the branch main

# Rebase c743dbd..d6c6a6a onto c743dbd (3 commands)
#
# Commands:
# p, pick <commit> = use commit
# r, reword <commit> = use commit, but edit the commit message
# e, edit <commit> = use commit, but stop for amending
# s, squash <commit> = use commit, but meld into previous commit
# f, fixup [-C | -c] <commit> = like "squash" but keep only the previous
#                    commit's log message, unless -C is used, in which case
#                    keep only this commit's message; -c is same as -C but
#                    opens the editor
# x, exec <command> = run command (the rest of the line) using shell
# b, break = stop here (continue rebase later with 'git rebase --continue')
# d, drop <commit> = remove commit
# l, label <label> = label current HEAD with a name
# t, reset <label> = reset HEAD to a label
# m, merge [-C <commit> | -c <commit>] <label> [# <oneline>]
#         create a merge commit using the original merge commit's
#         message (or the oneline, if no original merge commit was
#         specified); use -c <commit> to reword the commit message
# u, update-ref <ref> = track a placeholder for the <ref> to be updated
#                       to this position in the new commits. The <ref> is
#                       updated at the end of the rebase
#
# These lines can be re-ordered; they are executed from top to bottom.
#
# If you remove a line here THAT COMMIT WILL BE LOST.
#
# However, if you remove everything, the rebase will be aborted.
#
~
~
~
~
~
~
~
~
~
```

Step 3: Check the resul

```
Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/local-repo (main|REBASE 2/3)
$ git log --oneline
ed34a48 (HEAD) Revert "added file apple in main branch"
c743dbd added file apple in main branch
33bfb00 (origin/main, origin/HEAD) Added .gitignore in main baranch
6223876 (origin/Apple) Merge pull request #1 from muqeem9256-bot/Branch-git
11dae7c Create sample file in GitHub branch
cc25d15 Add test content to .ignore file
a7d09e1 Add initial README.md with test content

Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/local-repo (main|REBASE 2/3)
$
```

#COMPLETED.